Hardware Design File

SQRT (Sub-zero Quantitative Research Team)

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Key Reference Documents

1. UCD-DRD-2026-0
2. UCD-CDR-PMP-0
3. UCD-CDR-SDF-0

1 Executive Summary

This document outlines the hardware design for the SQRT TupperSat which investigates the flash evaporation and thermal evolution of liquid water under low-pressure, low-temperature conditions. The mission design is outlined with regards to overall design and justification in Section 3, including analysis of constraints and trade-offs in the design process.

Section 4 describes how the mechanical structure is optimised for strength and low mass to fit requirements. Section 5 shows how the electrical system integrates environmental, thermal, and positioning sensors, powered by a 5,000 mAh battery and supported by an 868 MHz antenna for telemetry transmission. The thermal design required to fulfil system requirements is shown in Section 6. Section 7 includes further notes on payload specific design.

Section 8 provides the hardware development plan that the team will follow to manufacture and assemble all components necessary to fulfil project requirements. Critical elements are also identified to take testing priority during development.

The SQRt team will follow the design outlined in this document to make a compact and robust TupperSat capable of completing SQRT mission goals.

2 Mission Overview

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to answer the question of how liquid plumes on icy moons thermally evolve.

This investigation into the thermal mechanisms behind cryovolcanic plumes will be done using a high-altitude demonstration. This will involve the sudden exposure of liquid water within the TupperSat to a colder, lower-pressure environment which will be generated by elevating the TupperSat system to an altitude of 20-25 km. At 25 km altitude, the average atmospheric pressure is about 25 mbar (2.5% of that at sea level) with a corresponding boiling point of water of approx 21°C. To perform this exposure, an opening mechanism consisting of a valve will expose a sample of water to a colder, less pressured environment within the TupperSat called the low pressure chamber.

3 Design Overview & Justification

3.1 Design Overview

3.1.1 Design concept

The initial design concept consists of a cubic satellite limited to a maximum of 2 liters in volume and 800 grams constraint. The dimensions and final mass are yet to be defined; they will be addressed in later stages of the process. Current planned dimensions, used for the simulation's sketch, are 16x14x8.6 cm, resulting in 2 liters.

Ventilation holes are included to prevent structural damage due to pressure differentials during ascent.

Outside, an antenna is installed through the Tupperware for communication purposes (to fulfil mission communication requirements and ensure that at least some science data is delivered, as TupperSat recovery is not granted). Lifting eyes used to connect Tuppersat to the balloon are connected to the inner structure via structural rods that support the payload from the base to the top.

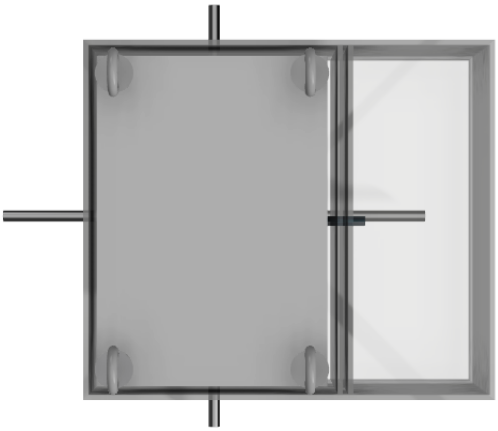


Figure 1: SQRT Initial Design — Top View

The internal display is divided into two sections for safety reasons, as the electronics can not be exposed

and affected by the liquid. The section division can be seen in Figure 1. One of the sections, called Electronics Bus (left section), will contain all the electronics, the actuator (servo motor and valve) and the liquid container. It will define the operation; record and save the thermal sensor's output; and send the data to ground, including the recorded data and the status of the valve. This Electronics Bus will be thermally insulated to maintain a proper operating temperature and the water in a liquid state.

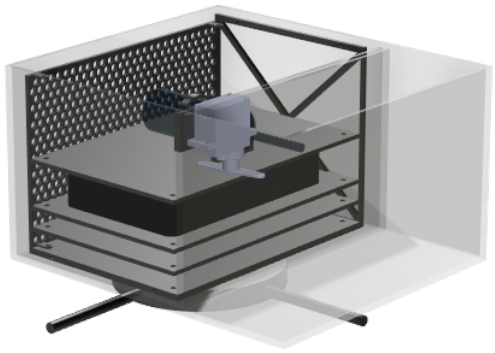


Figure 2: SQRT Mature Design Corner View

The other section, called the Low-Pressure Chamber (right section), will only house the thermal sensor. In this section, the water will be ejected and tested.

After a more mature design, it has been concluded that only one thermal sensor will be installed in the Low-Pressure Chamber to monitor the monitor the evolution of the water after valve opening. This thermal device will record the temperature on each pixel, so it can track the thermal evolution as pixel values or in picture format. It will capture thermal information throughout the entire flight to provide background and science frames.

Given the environment the liquid will be in, the most suitable design is considered a cylinder with a conical mouthpiece to fit a plastic tube to orient the ejection and locate the servo valve.

3.1.2 Valve Options for Internal Water Release

The following valve options were considered for controlled water release from the internal reservoir into the experiment chamber:

- **Pinch valve**
  A flexible tube is mechanically compressed to block the flow and released to allow water discharge. This solution provides excellent sealing, high tolerance to vibration and freezing, and keeps the actuator isolated from the fluid.
- **Servo-actuated ball valve**
  A quarter-turn ball valve is rotated by a servo motor to open or close the flow path. This option offers a large flow area and clear open/close states but requires higher actuation torque and careful sealing.
- **Rupture membrane (burst disk)**
  A thin membrane seals the outlet and is intentionally punctured or ruptured to release the water. This approach ensures zero leakage during ascent but allows only a single, non-reversible release event.

The pinch valve was selected as the preferred option due to its simple design and reliable sealing performance, which are well-suited to the vibration and rotation expected during ascent. Since the actuator

does not come into direct contact with the water, the risk of leakage is reduced, and the system is less sensitive to partial freezing or evaporation.

Inside pinch valve designs, most commercially available options are based on solenoid actuation. However, these valves are generally too heavy for the mission constraints, with typical masses of around 200g for the lightest models, and their performance can be unreliable at low temperatures due to increased material stiffness and resistance.

For this reason, a servo-actuated pinch valve was selected and implemented using a 3D-printed mechanism, providing a lightweight, mechanically robust, and more adaptable solution for the expected flight conditions.

The selected servo-actuated pinch valve is fully reversible and allows repeated open–close cycles, enabling extensive ground testing before flight. Furthermore, once the water is ejected, the valve can be actuated again to seal the nozzle and maintain a controlled internal configuration.

3.2 Design Justification

3.2.1 Spatial allocation

The final container dimensions are still to be defined; the dimensions used in the model are preliminary (16x14x8.6, length x width x height). Still, the design will be inside the thresholds established for mass and volume: 800 grams and 2 liters.

To divide the two chambers with enough space for the water to spread in the Low-Pressure Chamber and the electronics to be safe and fit with the water container, double 2mm plastic wall are considered, to ensure that no evaporated water reaches the electronics and compromises the mission. With these walls, the low-pressure chamber is 53 mm long.

Considering the optimal design, the liquid container is considered a syringe of 15 ml, with the pusher at the bottom fixed. The dimensions are 20x50mm (diameter x height) without the pusher structure. It will be placed in the highest floor of the structure, just above the battery, so the residual heat produced by the component reaches the syringe faster.

3.2.2 Water State Estimation

According to previous missions data, the temperature inside the Electronics Bus (insulated section) will be around 30ºC. At those altitudes (between 20 and 25 km), the temperature outside is around -50 °C and the pressure is between 50 and 25 mbar. For our mission, the water is considered to be ejected at a 23km height where the pressure is around 35 mbar, as it is not guaranteed that the balloon will reach 25km. According to the following diagram, the water will be in a liquid state in the container given the external pressure and internal temperature.

As the aim of the experiment is to observe a visible stream or spray during the release, the water needs to be at a higher temperature than the expected ambient conditions before the valve is opened. This is related to the behaviour of water under different temperature and pressure conditions, as shown in Figure 3.

When the liquid is exposed to the low-pressure environment at high altitude, the sudden pressure drop should cause part of the water to evaporate. If the initial water temperature is relatively low (around 10ºC), the water is more likely to freeze quickly, with limited vapour being produced.

The water container will need to be held at a temperature over 25ºC. To ensure this, the syringe is located as close as possible to the battery to absorb the emitted heat. In addition, bubble wrap will be installed all over the electronics bus to ensure that the heat and temperature are maintained, and isolate it from the outer cold.

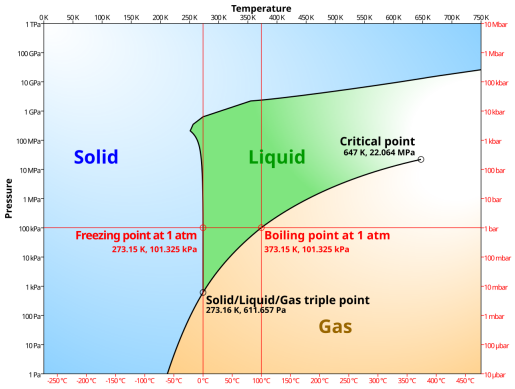


Figure 3: Simplified Phase Diagram for Water

4 Mechanical Design

This section will dive deeper into the design proposed in the design overview section. Once the basic design is proposed, the structure to hold the components and maintain integrity needs to be detailed.

The structure to hold the payload is based on an empty cube with three floors on the inside, to allocate and secure all the needed components. By adapting this solution, the design optimises the available space, using all the vertical space available, adding up to 22367.3 mm^2 for each of the three levels and base.

Each floor will be allocated in four different bolts, design inspired by the CubeSats from the lab.

Two different types of lateral panels are considered: the long ones are made of a honeycomb section, to add rigidity to the structure and absorb the impact in the landing, optimizing weight. The small lateral panels consist of a square with a supportive 'X' section to avoid buckling, and give access to the component and connection if needed.

4.1 Dimensions

The dimensions have been driven by two main constraints:

- **Height**
  The distance between each level shall be tall enough to accommodate all the components and their connections, but the shortest possible to use the space as efficiently as possible. The main components that have defined this are the servo motor and its valve.
- **Area**
  The battery is the biggest component, and so the base has to adapt to it. Furthermore, some margin needs to be left in each direction, so that there is space for the battery USB connection to feed the system. If the space is considered too long, it can be shortened.

The base and top are 125 mm in length and 90 mm in width, with 2 mm thickness to support and enclose the structure. They have 4x 2.5mm diameter holes drilled so that four bolts can go through it and connect with the Tupperware and give rails for the above levels, as well as connecting the large lateral panels. On the sides of both, there are 2x 1.5mm holes to connect the X lateral panels with short nuts. These holes are located at a safe distance from the 2.5mm ones to avoid internal cracks. Hole diameter and position can be modified if needed.

Each level is very similar to the base, with four holes drilled, but slightly smaller in width of 84 mm, as the large lateral panels are located on the base. The thickness is 2mm, as they don't need to support much effort. The thickness can always be modified if it is considered too thick or too thin.

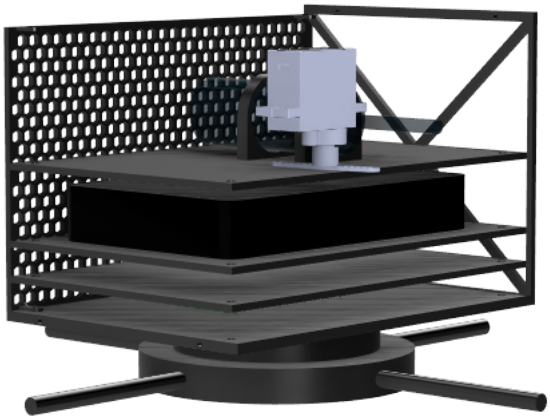


Figure 4: Close Structure View

The base will house the Raspberry Pi module, the sensors will be found in the first level. The second floor is only used for the battery, as it is the biggest component, and its USB cable connection. Lastly, the third level is for the syringe and the actuator (servo motor and pinch valve)

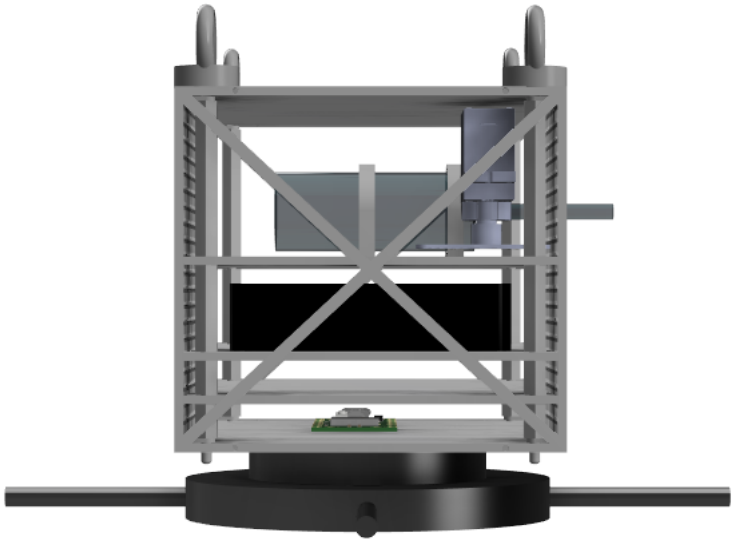


Figure 5: Lateral Structure View

Honeycomb panels are 125mm long, 80mm high and 2mm thick. They are mounted right above the base,

and under the top. Small appendices are located at the base and top to secure them to the structure.



Figure 6: Close View: Holes in the base, Levels and Appendices of the Honeycomb Walls

Figure 6 provides a better understanding of the holes display of the base and top (only base shown, but the same ones apply to the top), the alignment with the different levels, and the appendices to connect the honeycomb faces.

Lastly, the antenna is mounted under the base, with an extension to pass through the tupper frame. The antenna is composed of 4 metallic sticks in a cross display of 8.64 cm long each, with a gap of 1cm between the one on the same axis. The base is a cylinder of 92mm in diameter.

4.2 Floor division

Spacers will be used to divide the floors. The distance between each floor is as shown in Figure 7.

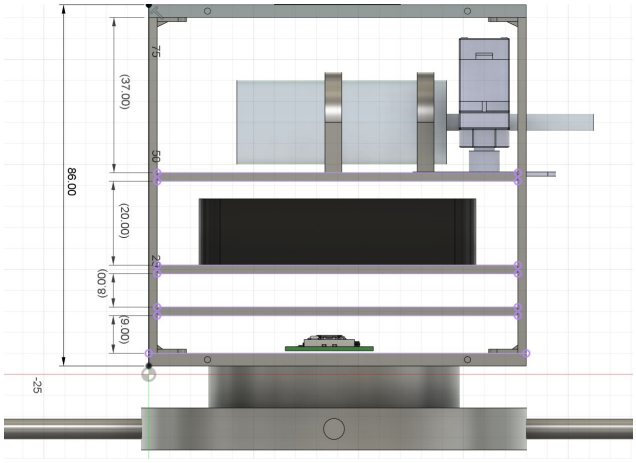


Figure 7: Side View Illustrating Structural Level Space Divisions

Figure 7 shows the divisions of levels within the electronics bus. The first space is 9mm. Here, it has to be considered that there are appendices from the honeycomb faces (1mm thickness), which are where the spacers will be located. Therefore, the real space is 8mm. A similar thing happens to the last space, 37mm turns into 36mm. With these measures, spacers of 4mm and 8mm can be used:

First space (8mm): 1x 8mm spacer.

Second space (8mm): 1x 8mm spacer.

Third space (20mm): 2x 8mm and 1x 4mm spacers.

Fourth space (36mm): 4x 8mm and 1x 4mm spacers.

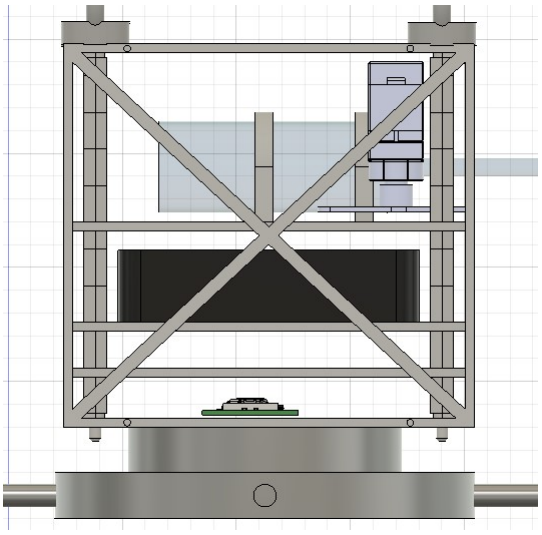


Figure 8: Side View of the Spacers Distribution to Define Inner Structure

The top will be secured using Eye nuts on each of the 4 bolts that go through the structure and tupper frame, closing and tightening the structure and floors. Bear in mind that the top will be placed over the appendices from the honeycomb faces (which lie on the spacers), so they will also be secured. Furthermore, the 'X' lateral faces will be attached to the base and top with bolts.

4.3 Mass Estimation

The mass estimate has been calculated by applying elastic PVC material to all components of the structure. It should be noted that this structure has one level, the first, which may not be necessary. This will happen if, during the assembly phase, its elements can be repositioned on the other levels, thus lightening the entire structure.

However, our calculations do not take into account the weight of the antenna, although based on previous designs, it will weigh around 80g.

As we can see in the following table, the structure fits very well with the estimates made in the PDR, being a bit over the 200g estimation. However, in the PDR the weight estimation fo the insulation material was 40g, but it will be lighter if using bubble wrap.

Table 1: Structure Weight Estimation

Component	**Estimated Weight (g)**
Top	29
Honeycomb	12 (x2)
Base	29
First and second level	28 (x2)
Syringe level	30
X Face	3.5 (x2)
8mm spacer	0.25 (x32)
4mm spacer	0.14 (x8)
Eye nuts	0.10 (x4)
Total weight	**224.12 g**

5 Electrical Design

The central control and processing unit of the designed electronic system is a Raspberry Pi module. Various sensors are connected to it using different communication protocols: DS18B20 for temperature sensing with 1-Wire communication, MS5611-01BA03 for barometric pressure measurement with an I2C bus, and Adafruit MLX90640 as a thermal sensor with an I2C bus. Additionally, a GPS module with UART communication for positioning data is also included. The system has an antenna connector for communication. All the components have a common ground and operate at 3.3 V or 5 V levels as per their requirements.

The system also includes a DFRobot DF9GMS servomotor that is controlled by a PWM signal from the Pico. The system architecture uses a combination of multiple protocols (I2C, UART, 1-Wire, and PWM) that allow for the simultaneous measurement of environmental, thermal, and location information. The design of the system is equivalent to a modular embedded system that can be used for environmental monitoring, advanced instrumentation, or mechatronics.

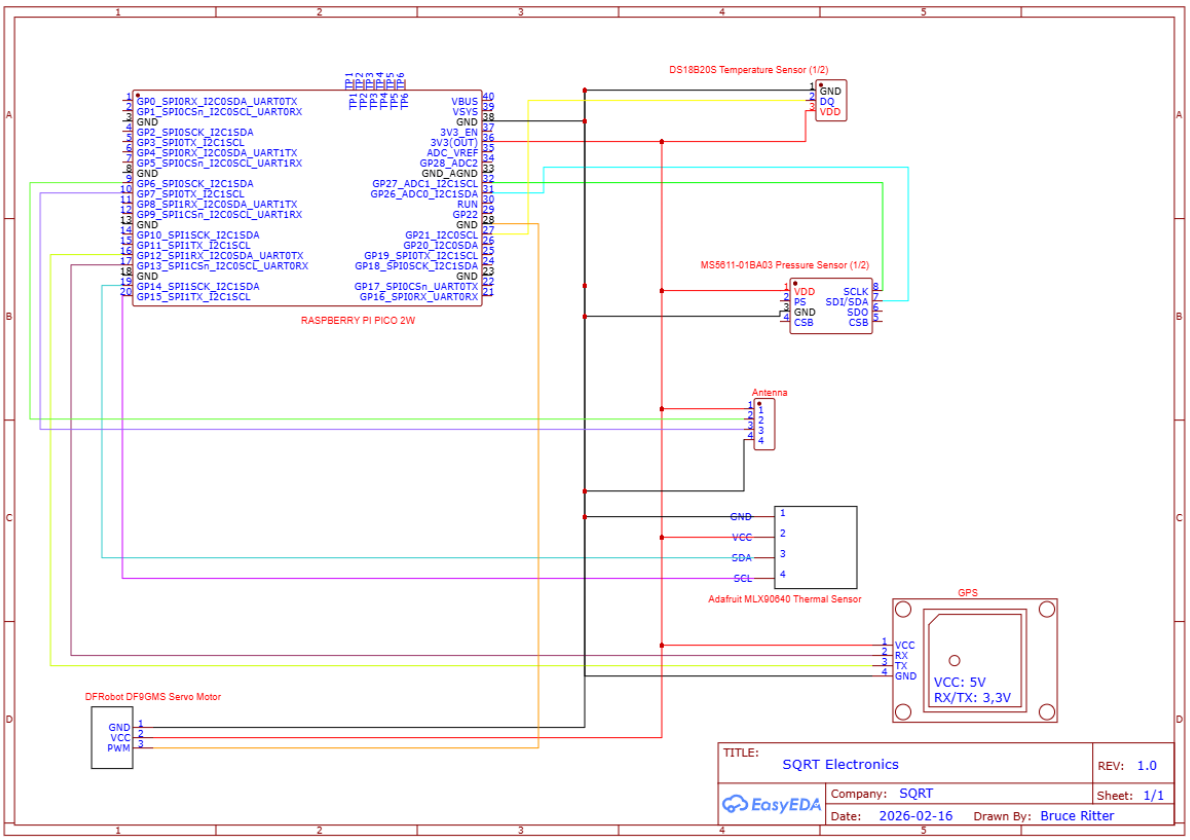


Figure 9: Electric circuit design

5.1 Power Source

This circuit will be powered by a Lithium Polymer battery technology, the PB-A5200 Power Bank. Thanks to its 5,000 mAh capacity, the battery is designed to power our circuit for up to 6 hours, which is enough time to cover the entire mission. It has a maximum current output of 2.1 A and an output voltage of 5V. The component will be connected via USB to the Raspberry Pi module.

5.2 Communication

To establish communication with the ground, an 868 MHz crossed dipole antenna will be used. This antenna consists of a 3D-printed 92mm cylindrical base and four 8.64 cm metal rods, with a 1 cm gap between them. It is located under the base of the structure to facilitate communication with the ground, as shown in Figure 10. The Tupperware will be modified to attach and expose the antenna to the structure.

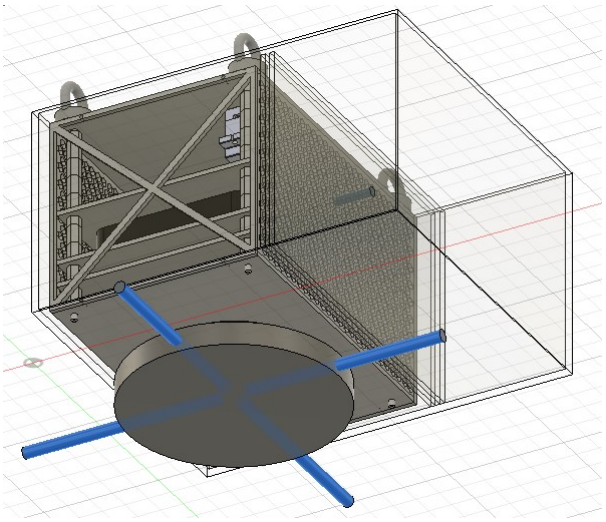


Figure 10: Antenna design

6 Thermal Design

Bubble wrap will be used as an insulation material due to its insulating properties and light weight. This will be placed around the entire structure in the gap between it and the Tupperware. In this way, the residual heat generated by the battery will be retained inside, allowing an optimal working temperature for the electrical components and keeping the water in a liquid state. In addition, the insulation will prevent this heat from passing into the low-pressure chamber, thus avoiding contamination of the experiment results.

7 Payload Design

The details of the payload design are provided in Section 4.

However, it is important to note that, although the design places most of the weight on one side of the Tupperware container (where the structure with the payload is located), and therefore its centre of mass, this does not present any problems. This is because this displacement has already been analysed in advance, and to avoid stabilization problems, the anchors to the balloon have been placed so that the centre of mass is inside the structure that houses the payload.

Therefore, the Tuppersat will be anchored to the balloon on the side with the most weight, thus maintaining the centre of mass at a point of equilibrium and avoiding unwanted swaying (beyond that produced by the ascent itself).

8 Hardware Development Plan

In future stages of the project, especially in the manufacturing, it can be confirmed that the first level is not needed, as the components can be relocated along the base and syringe level. This can not be stated yet, as the wiring and connections of the components may need more space than the modelling estimations if only using two internal levels.

For this reason, the first components that should be printed are the base and levels, so the component allocation could be tested rapidly. If occasionally one level is not needed, the model can be modified easily to readjust to the new optimized structure, avoiding delays. This redesign will induce a mass reduction.

The thermal sensor location is not fixed yet. Different positions will be tried during the assembly to search for the best location, to get the best view of the water jet thermal evolution.

During hardware design, testing of two components is recognised as critical:

- **Actuator and plastic nozzle**
  The mission relies on the actuation of the syringe; it needs to seal the syringe properly from the low-pressure chamber to avoid evaporation and spill. Furthermore, the plastic material has to be elastic enough to be deformed to the pinch position with the applied force, and avoid freezing in parts exposed to low pressure and temperature.
- **Insulation material**
  The water needs to be in a liquid state, or at least most of it. To achieve this, the bubble wrap thermal efficiency has to be tested to demonstrate that it can retain the heat produced from the battery in the Electronics Bus.